

# Aula Conectada

Investment case · Bolivia

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2026-06-19

# 1 · Diagnostic — gaps in education

# Target population and data sources

We draw on **two independent INE surveys**: **EH 2024** (education, access) and **EDSA 2023** (the home environment — violence, discipline). At the overlapping age band (15–19) they agree closely, so the two can be read together with confidence.

| Indicator — girls/women 15–19, weighted | EH 2024 | EDSA 2023 |
|-----------------------------------------|---------|-----------|
| Currently attending school              | 86.0%   | 84.8%     |
| Mean years of schooling                 | 9.95    | 10.0      |
| % rural                                 | 25.5%   | 26.3%     |
| % indigenous                            | 17.0%   | 16.2%     |
| Unweighted n (15–19)                    | 3,530   | 2,224     |

# Key takeaways



Bolivian girls are **already in school** (~95% enrolment, no gender gap in access). The binding problems lie in **what marginalised girls learn, whether they stay, and the home environment they learn from.**

## What the diagnosis shows

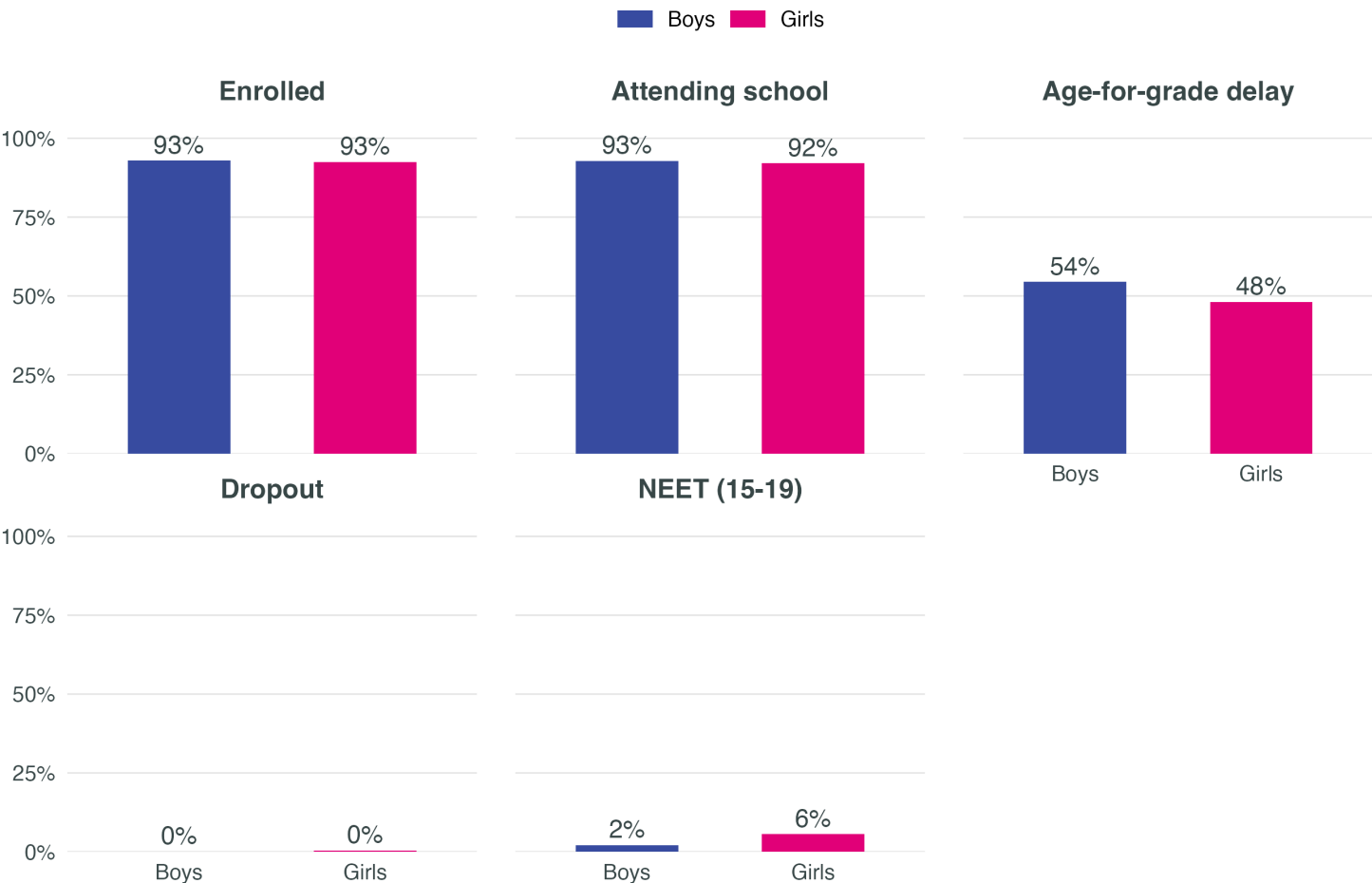
- The gap is **not access** — it is learning quality and staying in school, concentrated among **rural, indigenous and poor girls**.
- **Violence at home is part of the story for girls.** Violent discipline is concentrated among **rural-indigenous** girls (~59% vs 31% urban non-indigenous), and girls in these homes have **~2 fewer years of schooling**.
- **Early pregnancy** is the single strongest schooling correlate (**-2.9 years**), pointing to a second, distinct vulnerability for adolescent girls.

## What this implies for Aula Conectada

- Fund the **learning layer** — gender-responsive pedagogy and teacher training (Evans & Yuan, 73+ RCTs) — not hardware.
- Address the **home environment**: a low-cost **parenting** component for the violence pathway.
- **Flag** a girls'-empowerment component (ELA-style) for the early-pregnancy pathway — for discussion with policymakers.
- Target the **most vulnerable**: rural and indigenous girls.

# Gender gaps in education

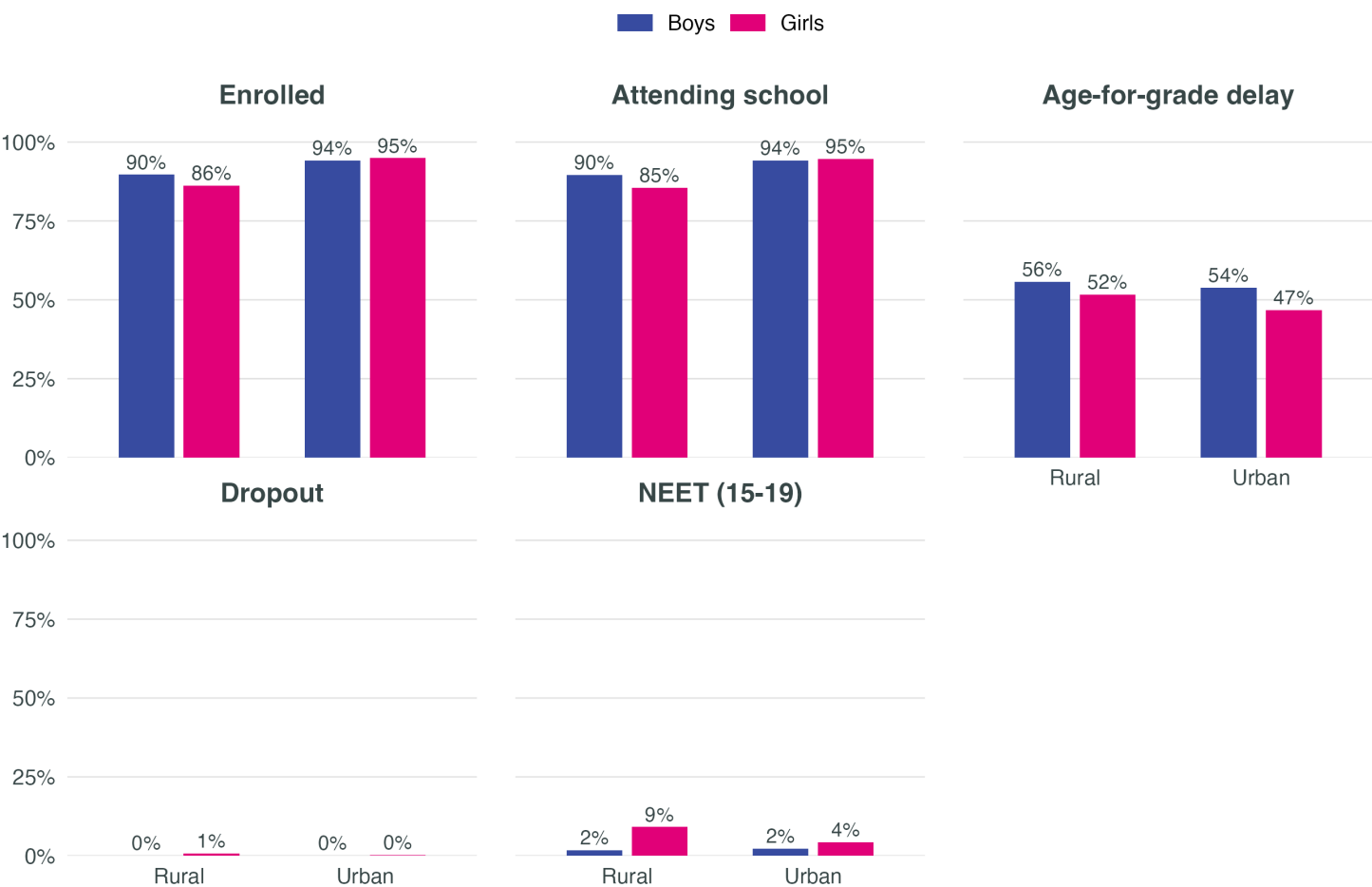
Adolescents 10–19 in Bolivia



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

# Education outcomes by area of residence

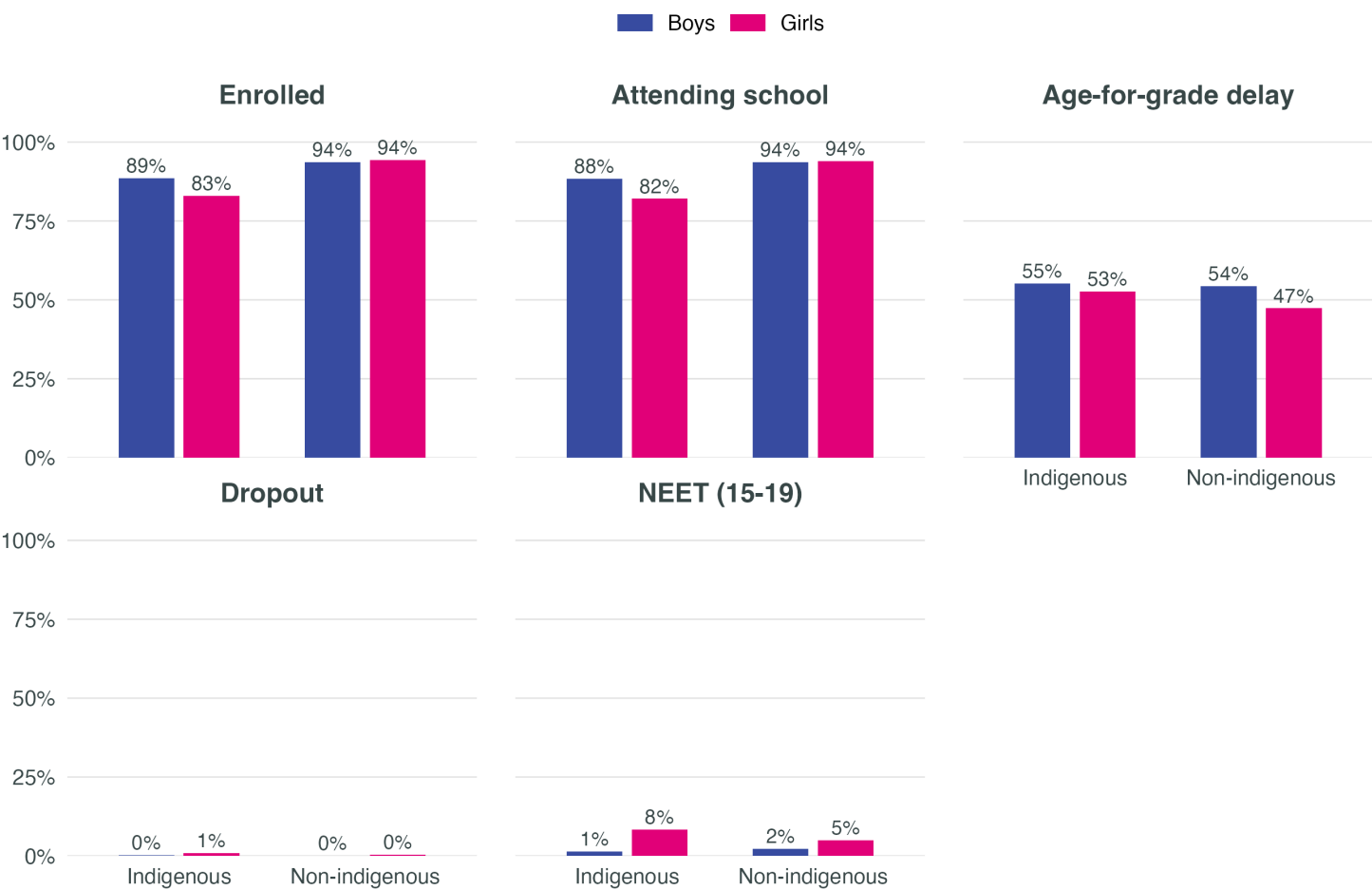
Adolescents 10–19, by sex and rural/urban area



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

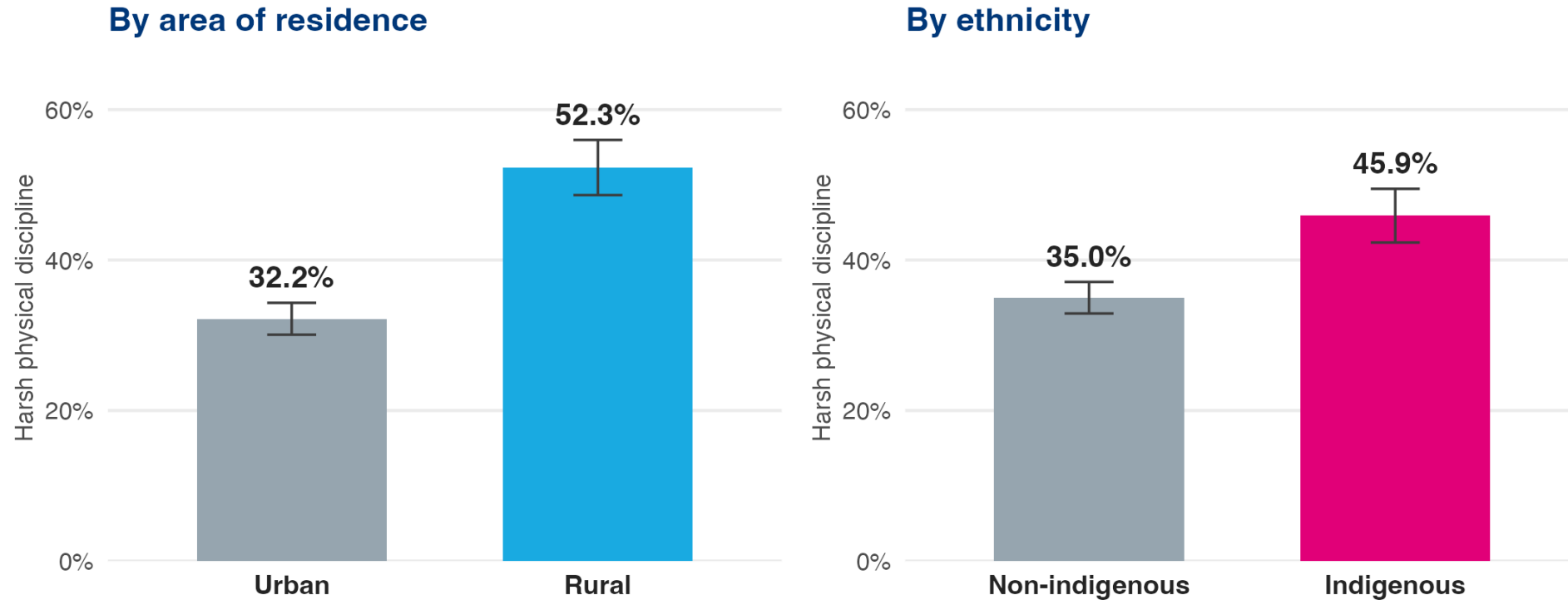
# Education outcomes by indigenous status

Adolescents 10–19, by sex



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

# The home environment: violent discipline is higher in rural and indigenous homes



EDSA 2023, INE Bolivia · survey-weighted · women with children · whiskers = 95% CI

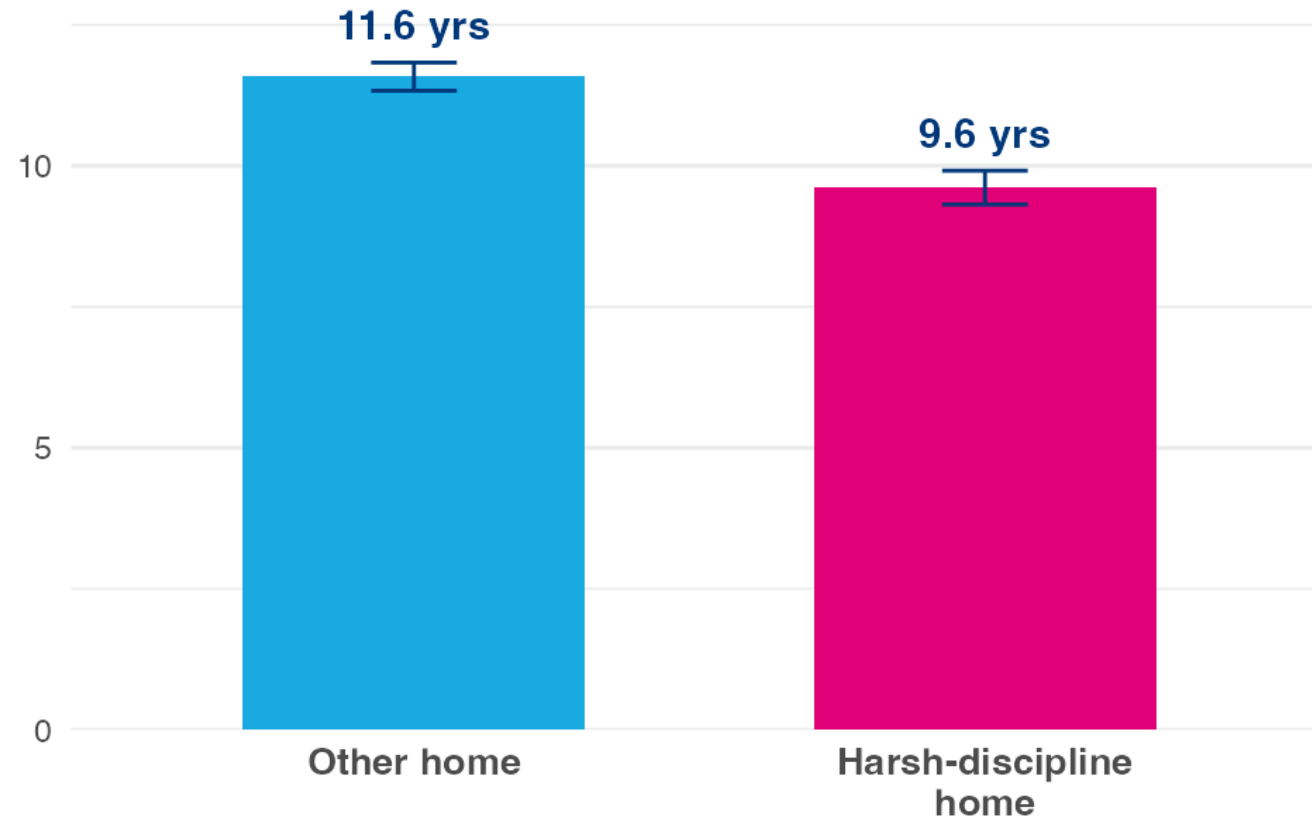
Harsh physical discipline of children (EDSA 2023, survey-weighted, women with children). Both axes raise risk **independently**: rural homes report 52.3% vs 32.2% urban (a ~20 pp gap), and indigenous homes 45.9% vs 35.0% non-indigenous (a ~11 pp gap). Rural location is the stronger axis, but indigeneity adds risk on its own — so targeting follows **rural + indigenous**, not a single dimension.



# Harsh discipline tracks lower schooling

## Years of schooling by household discipline

Women 12–49 · EDSA 2023, survey-weighted



EDSA 2023, survey-weighted. Women in harsh-discipline homes average ~2 fewer years of schooling than women in other homes. *Note:* the violence/discipline module in EDSA is asked of women only, so these figures reflect women's responses about their own households; comparable male-respondent data are not collected. This motivates the regression that follows.

# Violent discipline is associated with lower schooling

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| Years of schooling (women 12–49)  | Association | Std. error | p      |
|-----------------------------------|-------------|------------|--------|
| Harsh physical discipline in home | −0.90       | 0.16       | <0.001 |
| Indigenous                        | −1.10       | 0.20       | <0.001 |
| Rural                             | −0.05       | 0.32       | 0.87   |
| Wealth Q2 vs Q1                   | +1.42       | 0.25       | <0.001 |
| Wealth Q3 vs Q1                   | +2.74       | 0.37       | <0.001 |
| Wealth Q4 vs Q1                   | +4.49       | 0.35       | <0.001 |
| Wealth Q5 vs Q1                   | +6.69       | 0.35       | <0.001 |
| Age                               | −0.12       | 0.01       | <0.001 |

**Living in a harsh-discipline household is associated with ~0.9 fewer years of schooling**, holding wealth, ethnicity, area and age constant. Separately, **early pregnancy (<18) is associated with −2.9 years** (standalone model, n=391, p<0.001) — a second, distinct pathway.

## 2 · Evidence → Theory of Change

# From diagnosis to three matched responses

The diagnosis points to three distinct problems, each with a different evidence-based answer:

## Problem 1 — learning, not access

Enrolment is ~95%; the gaps are in learning and the digital/STEM transition.

- Isolated hardware rarely improves learning  
(Cristia et al., 2017)
- Tools work **with structured pedagogy + teacher training**  
(Gómez-Ruiz & Franco, 2023)
- Gender-responsive content lifts **STEM aspirations**  
(Outes-León et al., 2020)

→ Anchor: **Evans & Yuan (2022)**, 73+ RCTs.

## Problem 2 — staying in school

Retention falters for the most vulnerable girls; empowerment and safe spaces keep them enrolled.

- Girls' clubs **raise enrolment (+8.5 pp)**  
(Bandiera et al., 2018 — ELA)
  - Life-skills + mentoring tackle the drivers of drop-out, including early pregnancy
- Anchor: **Bandiera ELA**,  $\phi$ -adjusted to Bolivia.

## Problem 3 — the home environment

Violent discipline is concentrated among marginalised girls and tracks lower schooling.

- Parenting programmes **cut violence** against children (IRR 0.55)  
(Cluver et al., 2017)
  - **Low-cost smartphone delivery**  
(Janowski et al., 2025 — ParentApp)
- Anchor: **Cluver (effect) + Janowski (delivery)**.

Each problem maps to one layer: the **learning layer** (the costed BCR anchor), the **empowerment layer** (ELA — the attendance/retention answer,  $\phi$ -adjusted), and the **home-environment layer** (parenting, for the violence pathway). Hardware is an enabler, not the intervention.

# What we are building — evidence-matched layers

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- ① Learning layer pedagogy & teacher training ② Empowerment ELA-style clubs (staying in school) ③ Home environment parenting support (violence)

Girls stay, learn & transition Narrower gaps in learning & work

## 3 · Projections — Aula Conectada in Bolivia

# Direct anchors — digital education and gender norms

Programmes with measured effects **on the outcomes** Aula Conectada seeks to improve — one per layer:

| Layer            | Anchor                           | Country       | Measured effect                   | Used for                                  |
|------------------|----------------------------------|---------------|-----------------------------------|-------------------------------------------|
| Learning         | Evans & Yuan (2022), 73 RCTs     | Meta-analysis | +0.07 SD learning (girl-specific) | Learning benefit (in SD)                  |
| Empowerment      | Bandiera et al. (2018) — ELA     | Sierra Leone  | +8.5 pp enrolment                 | Attendance → earnings ( $\phi$ -adjusted) |
| Home environment | Cluver et al. (2017) — PLH Teens | South Africa  | IRR 0.55 on child abuse           | Violence↓ → attendance (chain)            |

Each anchor is matched to the outcome it actually measured (Route A — no cross-unit conversion). The **learning** effect stays in SD and is reported as a learning benefit, not converted to attendance. The **ELA** and **parenting** layers carry the attendance effects (in pp),  $\phi$ -adjusted to Bolivia. Delivery model for the parenting layer: Janowski et al. (2025) ParentApp — low-cost, smartphone-based. Full  $\phi$  logic and the CBA chain in the assumptions slide and appendix.

# Key assumptions

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## The five $\phi$ factors are claims, not estimates

- $\phi_{\text{pop}} = 1$  when the source and target are identical in age and % rural
- $\phi_{\text{baseline}} = 1$  when Bolivia has full headroom; less when the indicator is already close to its ceiling
- $\phi_{\text{geo}}$  is banded by Haversine distance (1.00/0.85/0.65/0.50)
- $\phi_{\text{econ}}$  is capped at [0.30, 1.00] for stability
- $\phi_{\text{delivery}}$  reflects government implementation versus NGO/research

## Robustness

- Multiplicative form is **conservative**: each  $\phi \leq 1$ , so the projection only attenuates the external effect, never amplifies
- The ELA anchor is the **most transported** (Sierra Leone, post-conflict, in-person); we show its BCR contribution across  $\phi_{\text{ELA}} \in [0.40, 0.70]$  rather than a single point
- Sensitivity to alternative weightings in the technical methods paper (in preparation)



# Synthesis — expected returns from Aula Conectada

Three layers, each anchored on the outcome its evidence **actually measured** — no cross-unit conversion.

| Layer                        | Source effect                  | $\phi$ (transport) | Bolivia effect | Channel                                             |
|------------------------------|--------------------------------|--------------------|----------------|-----------------------------------------------------|
| Learning (digital pedagogy)  | +0.07 SD (Evans & Yuan)        | $\approx 0.86$     | +0.06 SD       | reported in SD — <i>not</i> converted to attendance |
| Empowerment (ELA-style)      | +8.5 pp enrolment (Bandiera)   | 0.55               | +4.7 pp        | keeps girls in school → school-years → earnings     |
| Home environment (parenting) | IRR 0.55 × prevalence (Cluver) | 0.85               | +1.9 pp        | violence↓ → attendance → school-years → earnings    |

The  $\phi$  (transport) column is the *composite* attenuation factor applied to each source effect ( $\phi = \prod \phi_k$  over the five dimensions — population, baseline, geography, economy, delivery). The full per-factor breakdown for each anchor is in the projection output ([05\\_projections.R](#)); see the assumptions slide for the ELA  $\phi$  range.

The two **attendance** effects act through **different mechanisms** (empowerment/safe-spaces vs the home environment), so they are **added**:  $\approx 6.6$  pp combined, feeding the Mincer → earnings chain. The **learning** effect is reported separately in SD — we deliberately do **not** claim a digital-only attendance effect, since a device alone does not address why girls leave school.

# CBA assumptions — every number, sourced

| Parameter                          | Value             | Source                                                                                        |
|------------------------------------|-------------------|-----------------------------------------------------------------------------------------------|
| Mincer return per school-year      | 7.5%              | EH 2024, Bolivian women (A12)                                                                 |
| Female annual wage                 | \$3,500           | INE EH 2024 weighted median, age 25–54                                                        |
| Working life · discount rate       | 35 yrs · 3%       | LAC norm · World Bank Educ. default (2–5% sens.)                                              |
| <b>PV per attended school-year</b> | <b>\$3,520</b>    | project Mincer pipeline ( <b>08_bcr_estimation.R</b> ) — discounted lifetime earnings premium |
| pp → attended school-years         | +2.4pp ⇒ +0.10 yr | over ~4 secondary years                                                                       |
| ELA enrolment effect               | +8.5 pp           | Bandiera (2018), Sierra Leone                                                                 |
| $\phi$ (ELA transport)             | 0.40–0.70         | Sierra Leone→Bolivia (range shown)                                                            |
| Violence→attendance gap            | –13.1 pp          | EDSA 2023, women 12–24 — <b>assoc. treated as causal</b>                                      |
| Cluver effect                      | IRR 0.55 (≈45% ↓) | Cluver (2017), PLH Teens RCT                                                                  |
| Harsh-discipline prevalence        | 38%               | EDSA 2023, mothers                                                                            |
| $\phi$ (parenting transport)       | 0.85              | LMIC→Bolivia                                                                                  |
| ELA cost · Parenting cost          | \$26 · \$10       | <b>benchmarks</b> (ELA \$17–35; Janowski \$8–12), not Bolivia-costed                          |
| Learning bundle cost               | \$152/girl        | deck costing (A7)                                                                             |

**Parenting chain:**  $+1.9\text{pp} = 13.1 \times 0.45 \times 0.38 \times 0.85$ . **Combined attendance** = ELA + parenting (different mechanisms, additive). The two assumptions to strengthen with a Bolivia pilot: the violence→attendance causal step, and the ELA transport factor  $\phi$ .

## 4 · Costing

# Unit costs — assumptions and sources

We construct **three scenarios (low / central / high)** from documented regional benchmarks. All values are annual per girl, in USD 2024.

| Component                     | Low | Central | High | Source                                                                                                                                   |
|-------------------------------|-----|---------|------|------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware amortised            | 30  | 48      | 60   | Plan Ceibal Uruguay TCO: \$192/4yr = \$48/yr (ACM TechNews 2013); low = bulk procurement, high = harder-to-reach areas                   |
| School connectivity           | 12  | 20      | 30   | Plan Ceibal connectivity+fibre: ~\$20/child/yr; rural premium drives high scenario                                                       |
| Digital content & platform    | 5   | 10      | 15   | IDB EdTech reviews 2018-2022: licensing + curated content cost \$5-15/child/yr                                                           |
| Teacher training (annualised) | 8   | 15      | 25   | Bolivia secondary teacher salary \$716/mo (boliviainformacion.com 2024); 20 days PD over 2 years amortised across ~1000 girls per cohort |
| Pedagogical support and M&E   | 5   | 8       | 12   | Plan Ceibal admin: ~\$27/yr (OLPC News 2010); fraction allocated to gender-pedagogy support                                              |
| TOTAL per girl/year (USD)     | 60  | 101     | 142  | Sum of components                                                                                                                        |

**Notes:** (1) Bolivia teacher salaries 2024: ~Bs 3,206/mo primary (~\$462), Bs 4,947/mo secondary (~\$716); rural permanence bonuses could lower net cost in priority zones. (2) Figures exclude initial capital (electrical infrastructure, classrooms) — assumed pre-existing. (3) International cost references (e.g. Plan Ceibal ~\$75–100/child) are context for the hardware scenario only; our central case is the pedagogy bundle, not device procurement.

# Component costs used in the CBA

The three layers enter the cost-benefit analysis at the following per-beneficiary costs (USD 2024):

| Component                                     | Cost             | Basis                                                        |
|-----------------------------------------------|------------------|--------------------------------------------------------------|
| Learning bundle (pedagogy + teacher training) | \$152 / girl     | deck costing, Bolivia-anchored                               |
| ELA-style empowerment                         | \$26 / girl      | benchmark (\$17–35/yr clubs) — <i>not yet Bolivia-costed</i> |
| Parenting (digital, Janowski model)           | \$10 / caregiver | benchmark (\$8–12) — <i>not yet Bolivia-costed</i>           |

The two benchmark costs are reported transparently and feed the layered cost-effectiveness; a dedicated Bolivia costing exercise would refine them before any scale-up decision.

# Programme scale — girls reached

Aula Conectada as investment in **adolescent girls 10–19**, with three priority-coverage scenarios:

| Scenario                        | N girls  | % of total | Coverage                                  |
|---------------------------------|----------|------------|-------------------------------------------|
| Pilot (8 priority departments)  | ~80,000  | 8%         | Rural+indigenous+poor                     |
| Intermediate scale              | ~280,000 | 29%        | Broad priority subset                     |
| Full scale (all priority girls) | ~568,000 | 58%        | All girls with at least one vulnerability |
| Universal (all girls 10–19)     | ~977,000 | 100%       | National coverage                         |

**Population figures** derived from weighted EH 2024 (Table A1). The programme was designed for vulnerability-based targeting — we do not recommend universal coverage as it would dilute impact in households without digital access needs. **Central planning scenario** is full priority scale (~568,000 girls). The **parenting layer** would reach the *caregivers* of these girls through the same priority targeting, at the separate benchmark cost noted in the costing section — concentrated on the rural and indigenous households where violent discipline is most prevalent.

## 5 · ROI — Cost-benefit analysis

# Cost-effectiveness by layer

Each layer is anchored on the outcome its evidence measured. The **learning** layer's benefit is in SD (learning); the **ELA** and **parenting** layers deliver attendance in pp.

| Layer                              | Effect ( $\phi$ -adj.)   | Cost                | Cost per unit gain       |
|------------------------------------|--------------------------|---------------------|--------------------------|
| Empowerment (ELA)                  | +4.7 pp attendance       | \$26 / girl         | ~\$6 per +1 pp           |
| Parenting (violence↓)              | +1.9 pp attendance       | \$10 / caregiver    | ~\$5 per +1 pp           |
| <i>Learning (digital pedagogy)</i> | <i>+0.06 SD learning</i> | <i>\$152 / girl</i> | <i>~\$25 per 0.01 SD</i> |

ELA at  $\phi=0.55$ ; parenting via the Cluver→EDSA chain at  $\phi=0.85$ . The two attendance layers are highly cost-effective *per pp* because their benchmark costs are low. The learning layer's benefit is a learning-quality gain (in SD), reported in its own unit — **~\$25 buys 0.01 SD** ( $\$152 \div 6$ ) — so we never convert across units or double-count.



# Combined BCR — shown as a range, not a point

The headline depends most on how much the ELA effect transports from Sierra Leone. We show it across  $\phi$  rather than pick one number:

| $\phi$ (ELA transport) | ELA pp | Parenting pp | BCR: ELA only | BCR: ELA + parenting |
|------------------------|--------|--------------|---------------|----------------------|
| 0.40 (conservative)    | 3.4    | 1.9          | 2.8           | 4.1                  |
| 0.55 (central)         | 4.7    | 1.9          | 3.9           | 5.1                  |
| 0.70 (optimistic)      | 5.9    | 1.9          | 4.9           | 6.1                  |

**What the BCR columns count:** the **numerator** is the monetised *attendance*  $\rightarrow$  *earnings* benefit only — “ELA only” counts the ELA attendance effect; “ELA + parenting” adds the parenting attendance effect. The **learning layer’s +0.06 SD benefit is NOT in the numerator** (it stays in SD, reported separately), but the learning **cost (\$152) IS in the denominator** of both columns alongside the ELA/parenting costs. So these BCRs already pay for all three layers while banking only the attendance benefit — i.e. a **conservative floor**, not a learning+ELA+parenting total. At the most conservative ELA transport ( $\phi=0.40$ ), ELA alone roughly reproduces the original access-only BCR (~2.8); parenting adds ~1.3. The combined figure is **illustrative** — it rests on the violence $\rightarrow$ attendance causal assumption and the ELA  $\phi$ ; the robust result is the **per-layer cost-effectiveness** on the previous slide.

# Benefit value chain — combined attendance to earnings

How the two attendance effects become a present-value benefit per girl (central case,  $\phi_{\text{ELA}}=0.55$ ):

| Step                                          | Calculation                                                           | Result            |
|-----------------------------------------------|-----------------------------------------------------------------------|-------------------|
| 1 • ELA attendance                            | Bandiera +8.5 pp $\times \phi$ 0.55                                   | +4.7 pp           |
| 2 • Parenting attendance                      | 13.1 pp $\times$ 0.45 (Cluver) $\times$ 0.38 (prev.)<br>$\times$ 0.85 | +1.9 pp           |
| 3 • Combined (different mechanisms, additive) | 4.7 + 1.9                                                             | +6.6 pp           |
| 4 • School-years gained                       | +6.6 pp over ~4 secondary years                                       | +0.28 yr          |
| 5 • Earnings (PV)                             | +0.28 yr $\times$ \$3,520/school-year                                 | ~\$970            |
| = Benefit vs. cost                            | \$970 $\div$ (\$152+\$26+\$10)                                        | BCR $\approx$ 5.1 |

The **learning** layer's +0.06 SD benefit is *additional* and reported in SD — not included in this earnings figure, so the monetised BCR remains a floor. Violence→attendance treated as causal for the projection (flagged). All constants on the CBA assumptions slide.

# What the BCR does and does not count

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The BCR monetises **only** the attendance → earnings channel. It is the **floor**, not the ceiling.

## Monetised

- Attendance / retention gains
- Bolivia Mincer return (7.5%/yr)
- Lifetime labour earnings (35 yrs)

## Not monetised (*documented, would only raise the return*)

- Gender norms and agency
- Household violence reduction
- Parent–child communication
- Intergenerational effects

# Conclusions



The evidence points to a clear, gender-focused design: invest in **how marginalised girls learn** and in **the home environment they learn from** — not in hardware.

## Who is most vulnerable

- **Rural and indigenous girls:** the largest schooling gaps (rural 8.7 vs urban 11.8 years) and the highest exposure to **violent discipline** at home ( $\approx 59\%$  rural-indigenous vs  $31\%$ ).
- Girls in **violent-discipline homes:**  $\sim 2$  fewer years of schooling (association,  $p < 0.001$ ).
- Girls facing **early pregnancy:** the single strongest schooling correlate ( $-2.9$  years).

## What to fund

- **Learning layer** — gender-responsive pedagogy, teacher training, computational-thinking content for girls (Evans & Yuan, 73+ RCTs). Hardware is an enabler only.

## What to add and flag

- **Parenting component** (Cluver effect + Janowski low-cost digital delivery) for the violence pathway.
- **Flag** a girls'-empowerment component (ELA-style) for the early-pregnancy pathway — for discussion with policymakers on budget and focus.

## Expected effect and return

- Costed learning layer:  $\approx$  **\$152/girl**, central **BCR  $\approx 2.8$**  [0.8–5.6] through earnings alone — a conservative floor.
- Parenting layer adds low-cost reach to the home environment; broader benefits (safer homes, the next generation) are **documented but not monetised** — the true return is higher.

# 6 · Methodological Appendix

# A0 • Cost-benefit analysis assumptions

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- **Estimated costing.** Unit costs are projected from regional benchmarks (Plan Ceibal Uruguay, IDB EdTech reviews, Bolivia teacher salary scale). An official programme costing may refine the range — we recommend reviewing before scale-up decisions
- **Management and coordination costs.** Figures include administrative overhead (10-20%) but exclude in-country monitoring, inter-institutional coordination, and political/operational costs in remote areas. These could add 10-20% extra in rural scenarios
- **Benefits only via the education-labour channel.** We do not quantify benefits in agency, reproductive health, violence reduction, or domestic productivity — all potentially significant but harder to monetise defensibly
- **Mincer estimated from Bolivian data.** The return per year of schooling is estimated from EH 2024 (7.5%/year for women, 95% CI [6.6%, 8.4%]). The estimate does not incorporate the effect of schooling on the *probability* of employment — the total benefit is likely larger than projected here
- **3% discount rate** is the World Bank Education Sector standard; high sensitivity to discount rate (at 5% central BCR drops proportionally)
- **Analysis excludes positive externalities** on younger sisters, community norms, and intergenerational effects on future daughters — all documented in the literature but outside the scope of this conservative BCR

# A1 · Data and sample — both surveys

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**EH 2024 (education, access, Mincer)** — INE Bolivia · Nov–Dec 2024 · 12,718 households, 39,497 persons · analytical subsample **6,254 adolescents 10–19** · design `upm/estrato/factor`, `srvyr (lonely.psu="adjust")`. Key vars: sex `s01a_02`; age `s01a_03`; indigenous = childhood/spoken language codes 1–37, 70, 71; poverty `poor`; disability Washington Group  $\geq 3$ ; HH internet `s06a_19`. Weighted OLS, PSU-clustered SE (`fixest::feols`). Microdata: [anda.ine.gob.bo/catalog/163](https://anda.ine.gob.bo/catalog/163).

**EDSA 2023 (home environment: violence, discipline, agency)** — INE Bolivia DHS round · Sep–Dec 2023 · **14,545 women 12–49**; discipline module asked of **mothers (4,678)** · design `upm/estrato/ponderadorm`, Taylor linearization (`srvyr`). Key vars: harsh discipline `ms11_1147/1148`; IPV `ms11_1105/1108`; attendance `ms01_0114`; years `aestudio`; indigenous self-ID `ms01_0108`; wealth quintile `riqueza` (Vivienda, merged by `folio`). Microdata: [anda.ine.gob.bo/catalog/119](https://anda.ine.gob.bo/catalog/119).

**Why two surveys:** EH carries the education/earnings case; EDSA carries the home-environment diagnosis. They are independent and **not individually linkable** — so EDSA calibrates the diagnosis and the violence→attendance association, never merged into EH at the unit level. At 15–19 the two agree within ~1pp on attendance, schooling, % rural and % indigenous (slide 3), supporting joint use.

# A3 · Empirical strategy

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**Base model (LPM with department fixed effects):**

$$Y_{ihd} = \alpha + \beta_F F_i + \beta_R R_i + \beta_I I_i + \beta_D D_i + \beta_P P_i + \gamma X_{ihd} + \delta_d + \varepsilon_{ihd}$$

where  $F$  = Girl,  $R$  = Rural,  $I$  = Indigenous,  $D$  = Disability,  $P$  = Poor,  $X$  = controls (age).

**Intersectional model with two-way interactions (× Girl):**

$$Y_{ihd} = \alpha + \beta_F F_i + \gamma X_{ihd} + \sum_{k \in \{R, I, D, P\}} \theta_k (F_i \times k_i) + \delta_d + \varepsilon_{ihd}$$

Each  $\theta_k$  measures how much larger or smaller the gender gap is in that subgroup. For example,  $\theta_R$  captures whether the gender gap between rural girls and boys differs from the gap between urban girls and boys.

**Intersectional model with three-way interactions (robustness):**

$$Y_{ihd} = \alpha + \text{main effects} + 2\text{-way} + \theta_{F \cdot R \cdot I} (F_i \times R_i \times I_i) + \delta_d + \varepsilon_{ihd}$$

and analogous for  $F \times R \times P$  and  $F \times I \times P$ . Three-way interactions capture whether the gender gap in a compound intersection (e.g. rural indigenous girls) differs from what the two-way interactions would suggest separately.

Estimation by weighted OLS using **factor** (expansion factor); standard errors clustered at PSU level (**fixest::feols**).



# A4 · Full coefficients table — main model

| Variable          | HH internet | Enrolled  | Attending | Any HH device | Age-for-grade delay |
|-------------------|-------------|-----------|-----------|---------------|---------------------|
| Girl (1=yes)      | 0.013       | -0.001    | -0.005    | 0.004         | -0.065***           |
|                   | (0.011)     | (0.008)   | (0.008)   | (0.005)       | (0.014)             |
| Rural             | -0.241***   | -0.068*** | -0.070*** | -0.044**      | 0.049**             |
|                   | (0.039)     | (0.020)   | (0.020)   | (0.022)       | (0.021)             |
| Poor (INE)        | -0.142***   | 0.012     | 0.008     | -0.016*       | 0.058***            |
|                   | (0.022)     | (0.008)   | (0.009)   | (0.008)       | (0.016)             |
| Indigenous        | -0.007      | -0.061*** | -0.063*** | -0.004        | 0.018               |
|                   | (0.036)     | (0.019)   | (0.019)   | (0.016)       | (0.024)             |
| Disability (WG3+) | -0.012      | -0.472*** | -0.469*** | 0.039***      | 0.356***            |
|                   | (0.069)     | (0.066)   | (0.066)   | (0.011)       | (0.052)             |
| N                 | 7,517       | 7,517     | 7,517     | 7,517         | 7,517               |
| R <sup>2</sup>    | 0.150       | 0.141     | 0.140     | 0.034         | 0.058               |
| Dept FE           | Yes         | Yes       | Yes       | Yes           | Yes                 |
| Weights           | Yes         | Yes       | Yes       | Yes           | Yes                 |
| Cluster SE (PSU)  | Yes         | Yes       | Yes       | Yes           | Yes                 |

Coefficients in decimal points (× 100 for pp). Robust SE in parentheses. \* p<0.10, \*\* p<0.05, \*\*\* p<0.01.

# A5 • Full intersectional table

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| Interaction       | Attending | Enrolled | HH internet | Any HH device |
|-------------------|-----------|----------|-------------|---------------|
| Girl × Rural      | -0.047*** | -0.043** | 0.024       | -0.008        |
|                   | (0.018)   | (0.018)  | (0.030)     | (0.013)       |
| Girl × Indigenous | -0.056**  | -0.052** | 0.013       | -0.006        |
|                   | (0.022)   | (0.023)  | (0.037)     | (0.013)       |
| Girl × Disability | -0.032    | -0.037   | -0.050      | -0.014        |
|                   | (0.137)   | (0.137)  | (0.132)     | (0.019)       |
| Girl × Poor       | -0.026    | -0.020   | 0.025       | -0.013        |
|                   | (0.016)   | (0.016)  | (0.028)     | (0.012)       |
| N (each model)    | 7,517     | 7,517    | 7,517       | 7,517         |

Each row is the coefficient on the (Girl × subgroup) interaction in a separate regression. Coefficients in decimal points. Standard errors in parentheses. Robust SE clustered at PSU.

# A7 · Formal foundation of the transportability approach

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We assume that the mismatch between the source population  $S$  (the anchor) and the target  $T$  (Bolivia) decomposes into  $K$  observable dimensions, with multiplicative attenuation:

$$\hat{\beta}_T \approx \hat{\beta}_S \cdot \prod_{k=1}^K \phi_k(S, T), \quad \phi_k \in [0, 1]$$

## Glossary of $\phi_k$ factors:

- $\phi_{\text{pop}}$  — standardised age and % rural
- $\phi_{\text{baseline}}$  — outcome headroom
- $\phi_{\text{geo}}$  — Haversine distance from La Paz
- $\phi_{\text{econ}}$  — GDP per capita ratio
- $\phi_{\text{delivery}}$  — 1.0 (government) or 0.7 (NGO/research)

## Assumptions:

- (A1) **Sufficient observables**: the 5 dimensions capture the relevant effect modifiers
- (A2) **Separability**: effects are approximately multiplicative
- (A3) **Monotonicity**: mismatch only attenuates, never amplifies
- (A4) **SE independence**: anchor variance propagates without perturbing the  $\phi_k$

# A8 · Bolivia-adjusted projection methodology

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Algorithm for each anchor:

1. Compute  $\phi_{\text{pop}}$  as  $1 - \sqrt{d_{\text{age}}^2, d_{\text{rural}}^2}$ , where  $d$  are standardised differences between source and Bolivia
2. Compute  $\phi_{\text{baseline}} = (Y_{\text{max}} - Y_{\text{Bol}})/(Y_{\text{max}} - Y_{\text{anchor}})$  when measurable (otherwise 1)
3. Compute  $\phi_{\text{geo}}$  in Haversine bands from La Paz:  $\leq 1,000$  km  $\rightarrow$  1.00; 1,000–5,000 km  $\rightarrow$  0.85; 5,000–10,000 km  $\rightarrow$  0.65;  $> 10,000$  km  $\rightarrow$  0.50
4. Compute  $\phi_{\text{econ}}$  as the Bolivia/anchor GDP per capita ratio, capped at [0.30, 1.00]
5. Compute  $\phi_{\text{delivery}}$ : 1.0 if government implementer; 0.7 if NGO/research (Vivalt 2020)
6. Multiply:  $\hat{\beta}_T = \hat{\beta}_S \cdot \prod \phi_k$
7. Propagate SE:  $\text{SE}(\hat{\beta}_T) = \text{SE}(\hat{\beta}_S) \cdot \prod \phi_k$
8. 95% CI:  $\hat{\beta}_T \pm 1.96 \cdot \text{SE}(\hat{\beta}_T)$

**Exclusion rules** — age overlap <25% with 10–19, or <2 demographic variables comparable  $\rightarrow$  anchor not projected.

**Uncertainty** — 95% CI propagated from the anchor SE. No synthetic noise on  $\phi$  factors.

# A9 • Consolidated evidence — anchors in use

| Programme                                  | Country       | Age   | Effect                  | Implementer  | ToC arrow         |
|--------------------------------------------|---------------|-------|-------------------------|--------------|-------------------|
| Girls' education meta-analysis (Evans &... | LMIC (pooled) | 10-19 | 0.070 (0.025) SD        | Government   | Digital education |
| Growth Mindset (Outes & Sanchez, full)     | Perú          | 12-14 | 0.054 (0.030) SD        | Government   | Digital education |
| Growth Mindset (Outes & Sanchez, regional) | Perú          | 12-14 | 0.135 (0.051) SD        | Government   | Digital education |
| Growth Mindset (Porter et al.)             | Sudáfrica     | 13-16 | -0.040 (0.100) SD       | NGO/Research | Digital education |
| ELA (Bandiera et al., 12-17 cohort)        | Sierra Leona  | 12-17 | 7.250 (2.700) score_pts | NGO/Research | Gender norms      |
| Irûmi — pensamiento computacional (Näsl... | Paraguay      | 7-8   | 0.089 (0.052) SD        | Government   | Digital education |
| Plan Ceibal SOLO HARDWARE (De Melo)        | Uruguay       | 8-12  | 0.000 (0.050) SD        | Government   | Digital education |
| Plan Ceibal + programación (Gómez-Ruiz)    | Uruguay       | 14-17 | 0.130 (0.050) SD        | Government   | Digital education |
| ProJoven (Ñopo et al.)                     | Perú          | 16-25 | 0.152 (0.060) ppts      | Government   | Labour market     |
| ParentApp Tanzania (Janowski 2025)         | Tanzania      | 0-17  | 0.130 (0.060) SD        | NGO/Research | Parenting         |
| Parenting for Lifelong Health (Cluver e... | Sudáfrica     | 10-17 | 0.100 (0.050) ppts      | NGO/Research | Parenting         |

Complete list of external anchors used in the Bolivia projection (see [05\\_projections.R](#) and [R/anchors\\_demographics.R](#)). The full systematic review of 52 papers is in [data/lit\\_review\\_papers.csv](#). The **ToC arrow** column indicates which arc of the Theory of Change each anchor feeds.

# A10 • Robustness — three-way interactions (OLS)

To test whether vulnerabilities compound, we estimate three-way interactions with OLS. If belonging to several vulnerable groups at once had a super-additive effect, the three-way coefficients would be significant for the outcomes with greatest heterogeneity.

| Three-way interaction     | Attending school | Enrolled       | HH internet    |
|---------------------------|------------------|----------------|----------------|
| Girl × Rural × Indigenous | -0.061 (0.045)   | -0.052 (0.046) | 0.050 (0.073)  |
| Girl × Rural × Poor       | -0.003 (0.036)   | -0.008 (0.038) | -0.096 (0.069) |
| Girl × Indigenous × Poor  | -0.002 (0.050)   | 0.005 (0.051)  | 0.096 (0.077)  |

**How to read:** each cell shows the three-way interaction coefficient and its standard error in parentheses. Significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Where no asterisks appear, the coefficient is not statistically distinguishable from zero at conventional levels. **Interpretation:** in this sample, the three-way interactions are not significant, indicating that the compound effect of belonging to multiple vulnerable groups (e.g. rural indigenous girl) is not substantially larger than the sum of effects of each group separately. Gaps accumulate roughly additively. This suggests well-targeted interventions on any axis (rural, indigenous, poverty) will reach girls who fit those characteristics.

# A12 · Bolivia Mincer estimation

| Spec | Term       | Coef       | SE       |
|------|------------|------------|----------|
| A    | aestudio   | 0.0559***  | (0.0034) |
| A    | exp_yrs    | 0.0376***  | (0.0031) |
| A    | exp_sq     | -0.0007*** | (0.0001) |
| A    | female     | -0.3257*** | (0.0177) |
| A    | rural      | -0.4146*** | (0.0484) |
| A    | indigenous | -0.0101    | (0.0227) |
| C    | aestudio   | 0.0747***  | (0.0046) |
| C    | exp_yrs    | 0.0325***  | (0.0041) |
| C    | exp_sq     | -0.0004*** | (0.0001) |
| C    | rural      | -0.3814*** | (0.0700) |
| C    | indigenous | -0.0218    | (0.0313) |

**Headline (Spec C, female-only):** each additional year of schooling raises expected log monthly labour income by 7.5% [95% CI: 6.6% – 8.4%]. This enters the BCR directly as the Mincer return.

**Sample:** 13,009 adults 25–60 with positive labour income (5,327 women · 2,488 rural). Weighted OLS, PSU-clustered SE. Model:  $\log(\text{ylab}) \sim \text{years\_edu} + \text{exp} + \text{exp}^2 + \text{female} + \text{rural} + \text{indigenous}$ .

**Caveats:** (i) OLS on the employed sample suffers from selection bias — the return may be over-estimated. (ii) We do not capture schooling's effect on the *probability* of employment — this under-estimates total benefit.

**Gender pay gap:** the *female* coefficient (Spec A) is the conditional gender wage gap — Bolivian women earn less than men with the *same* schooling and experience.

# A13 · Education outcomes by age band — overall

Early (10-14) vs late (15-19) adolescence



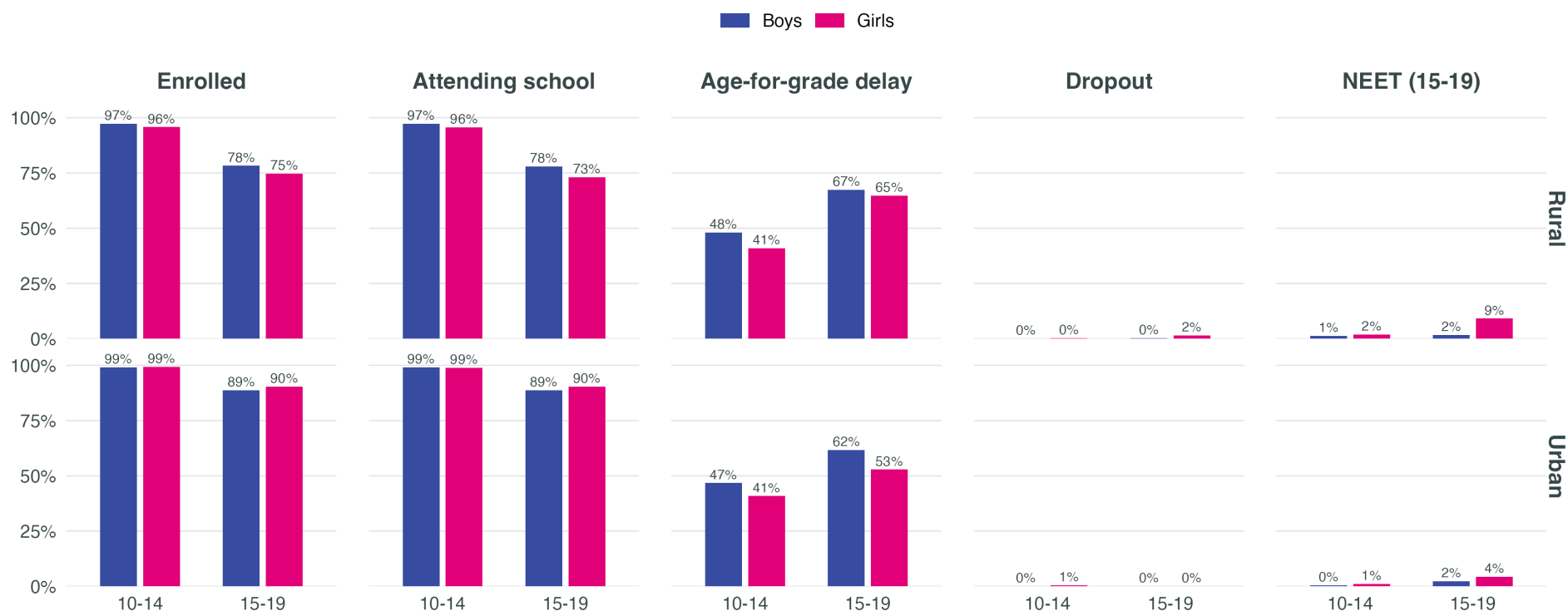
Source: INE Bolivia – Household Survey 2024. Weighted estimates.

NEET is only informative for 15-19 (in 10-14 nearly everyone is enrolled by construction).



# A14 · Education outcomes by age band — area of residence

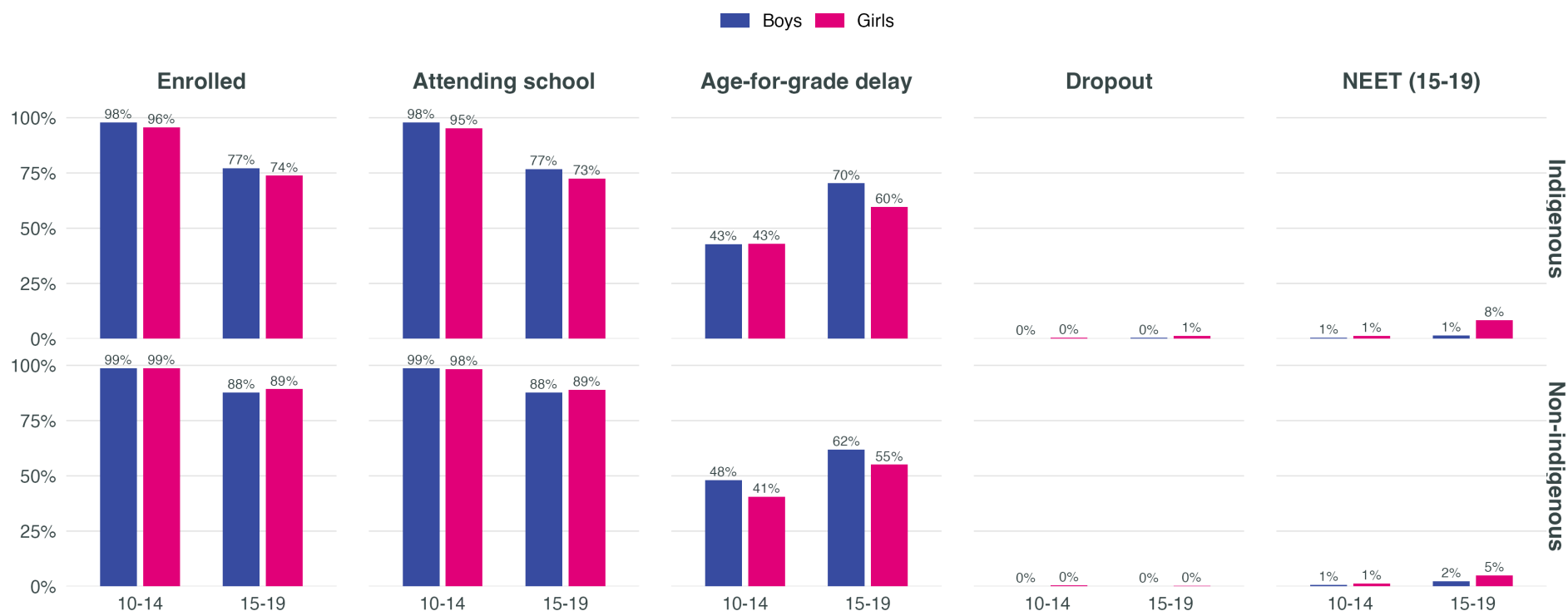
By sex, age (10-14 / 15-19) and area of residence



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

# A15 · Education outcomes by age band — indigenous status

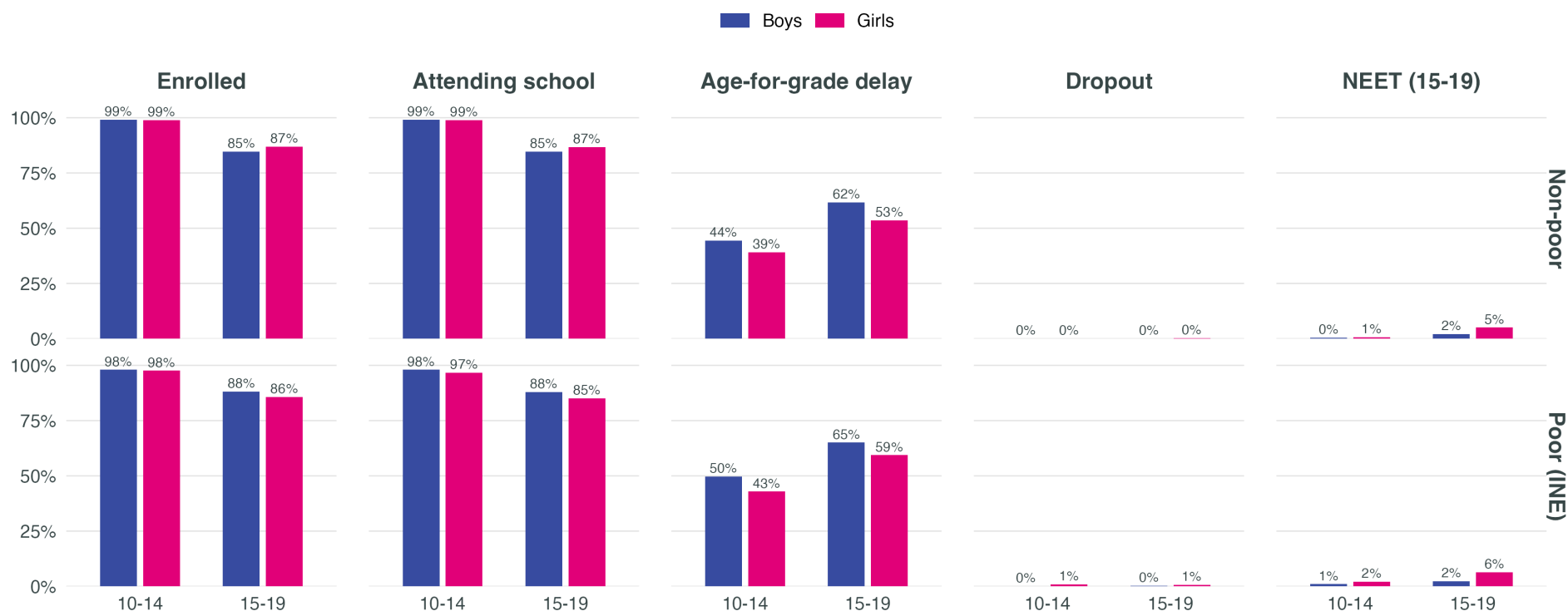
By sex, age (10-14 / 15-19) and indigenous status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

# A16 · Education outcomes by age band — poverty status

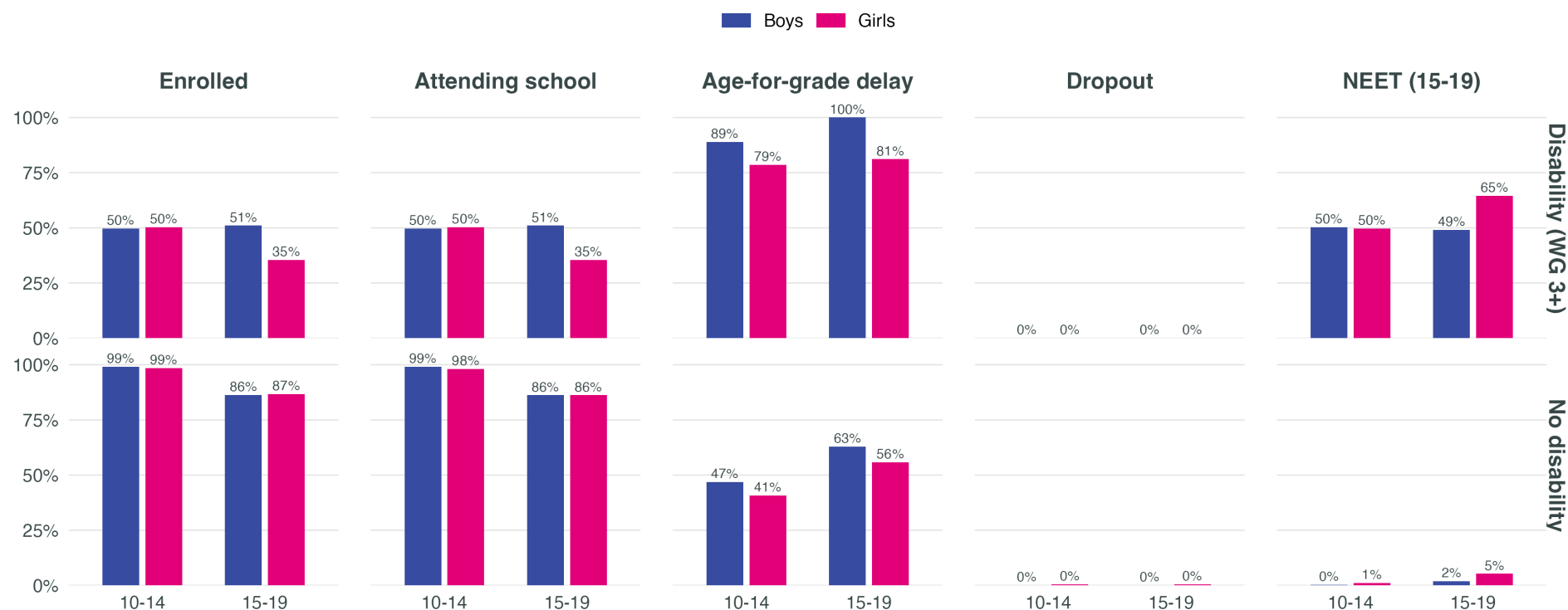
By sex, age (10-14 / 15-19) and poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

# A17 · Education outcomes by age band — disability

By sex, age (10-14 / 15-19) and disability



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

Disability subsample is small (~0.4%) — interpret with caution.

# A18 · Gender gaps by adolescent stage

Early (10-14) vs late (15-19) adolescence

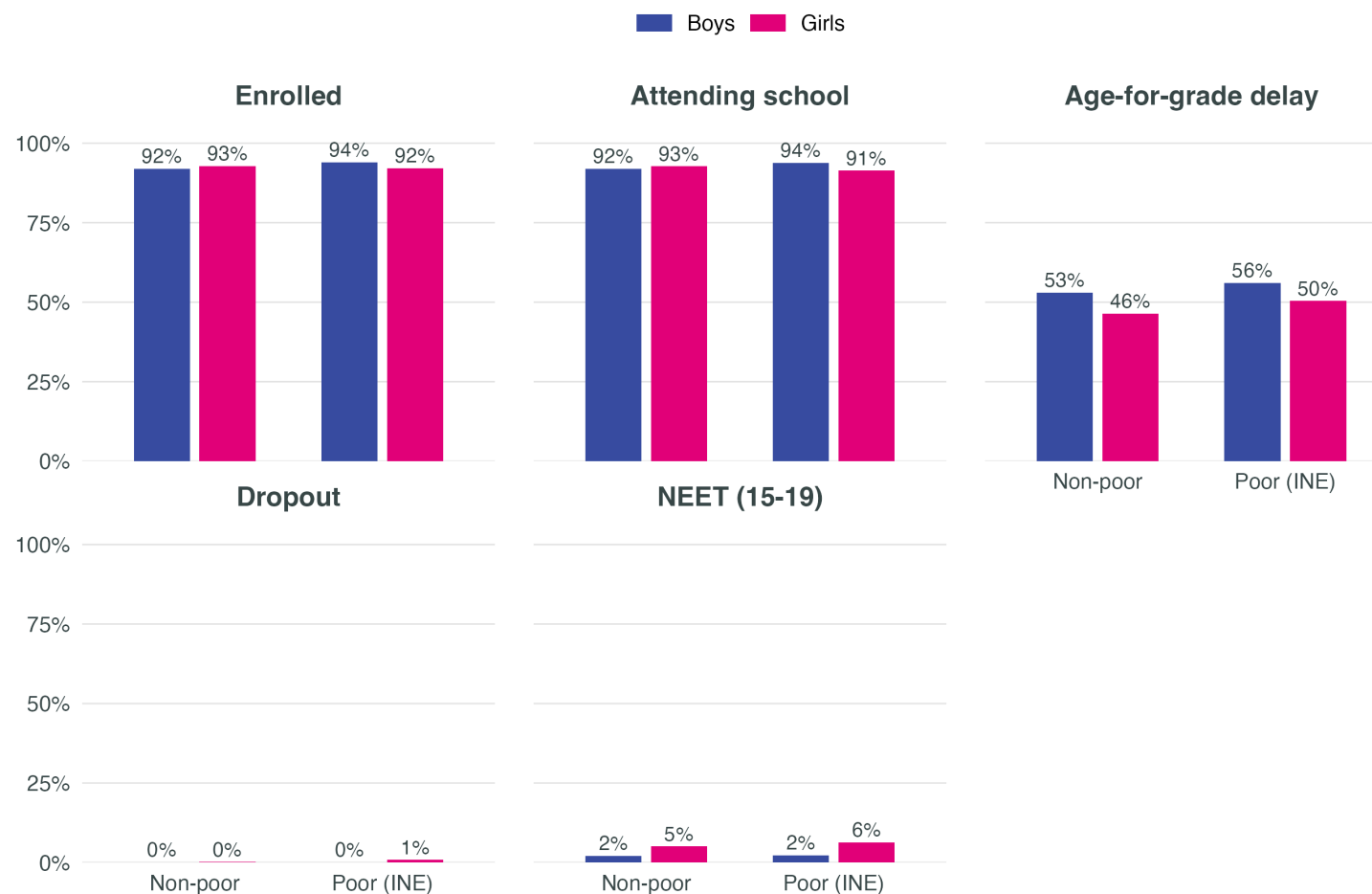


Source: INE Bolivia – Household Survey 2024. Weighted estimates.

NEET is only informative for 15-19 (in 10-14 most are in school by construction).

# A19 · Education outcomes by poverty status

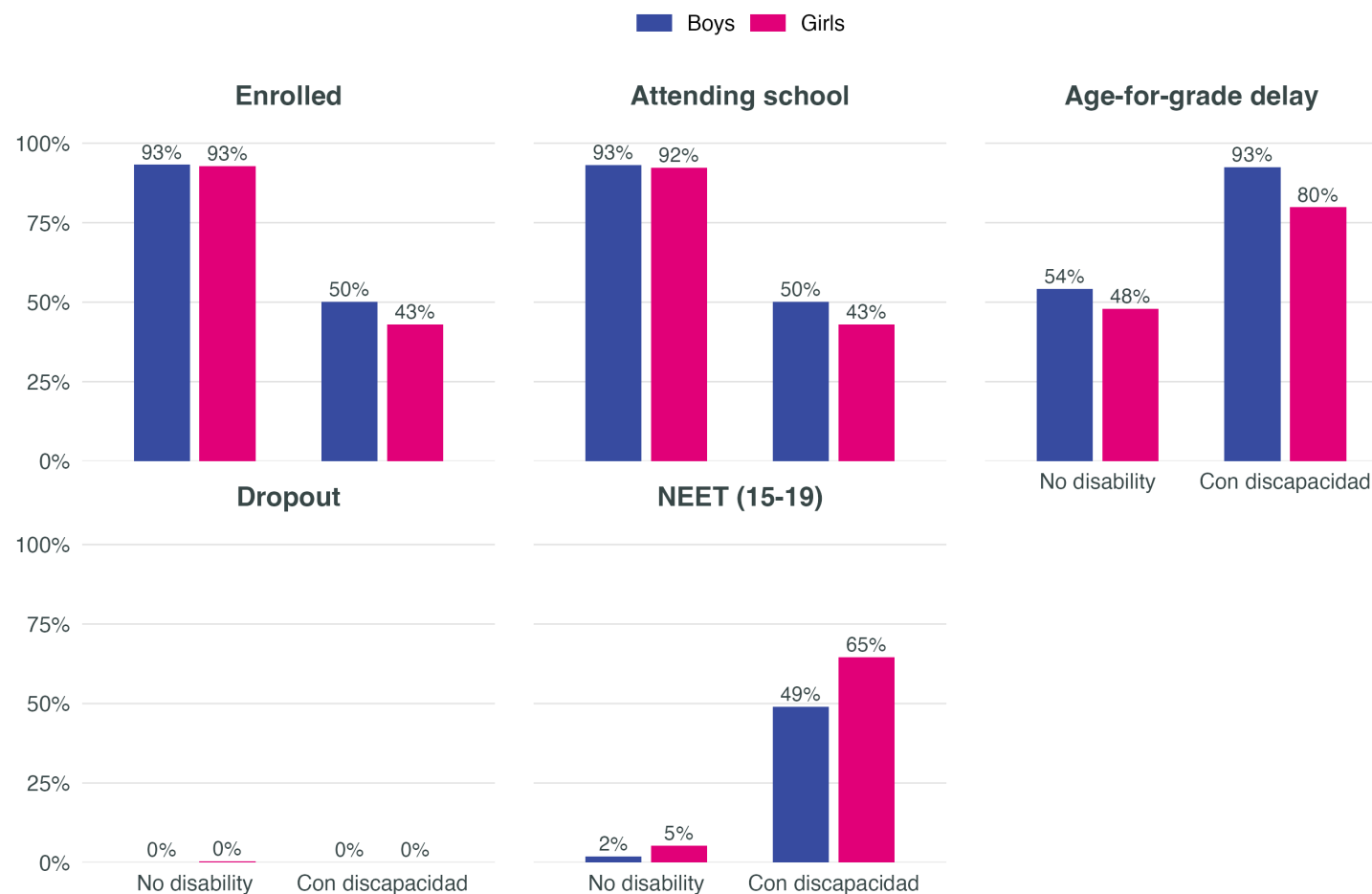
Adolescents 10–19, by sex and poverty status (INE)



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

# A20 · Education outcomes by disability

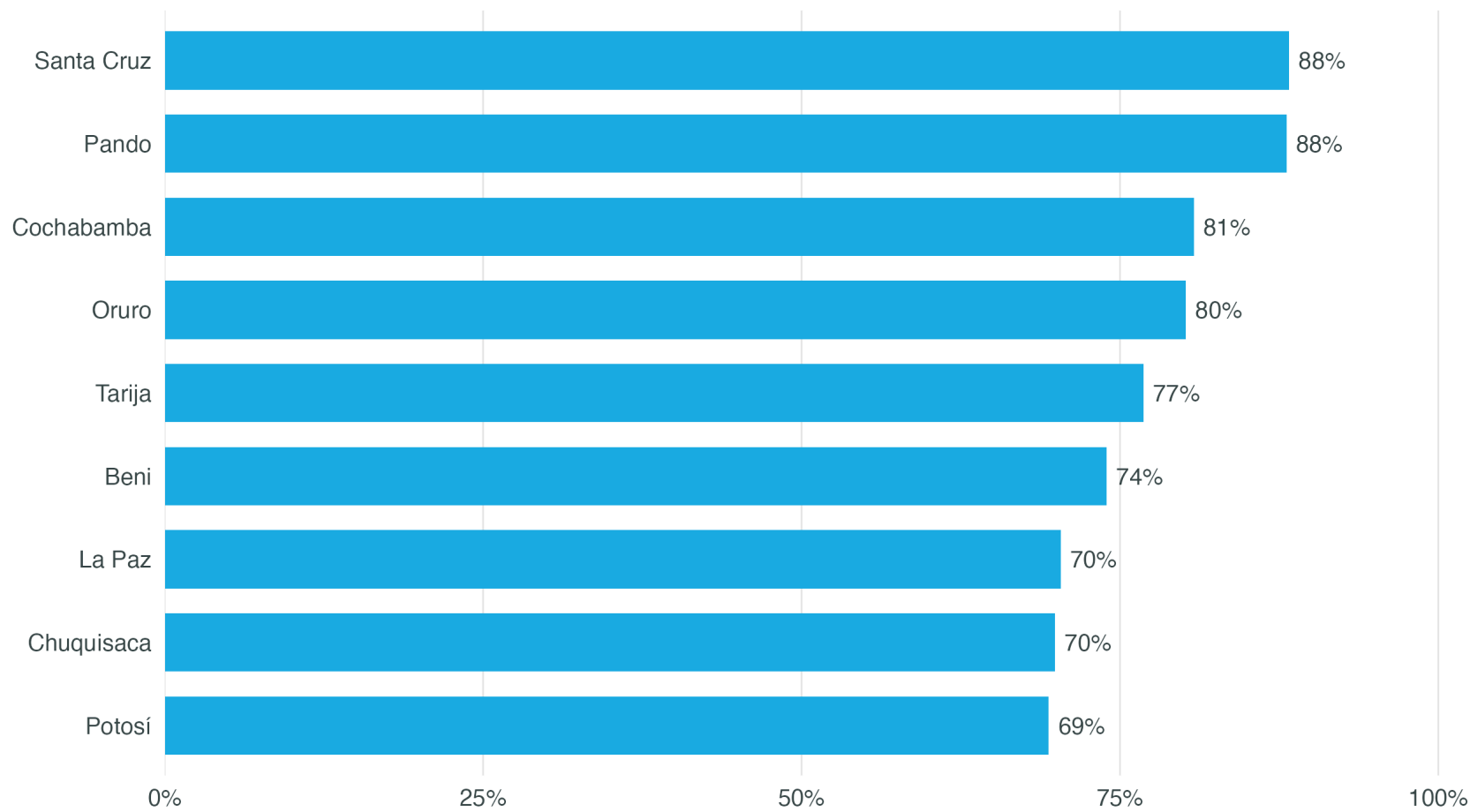
Adolescents 10–19, by sex · Washington Group severidad  $\geq 3$



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. Small sample size (~0.4%) — interpret with caution.

# A21 · Digital divide — departmental variation

Household internet, adolescent girls 10–19, by department

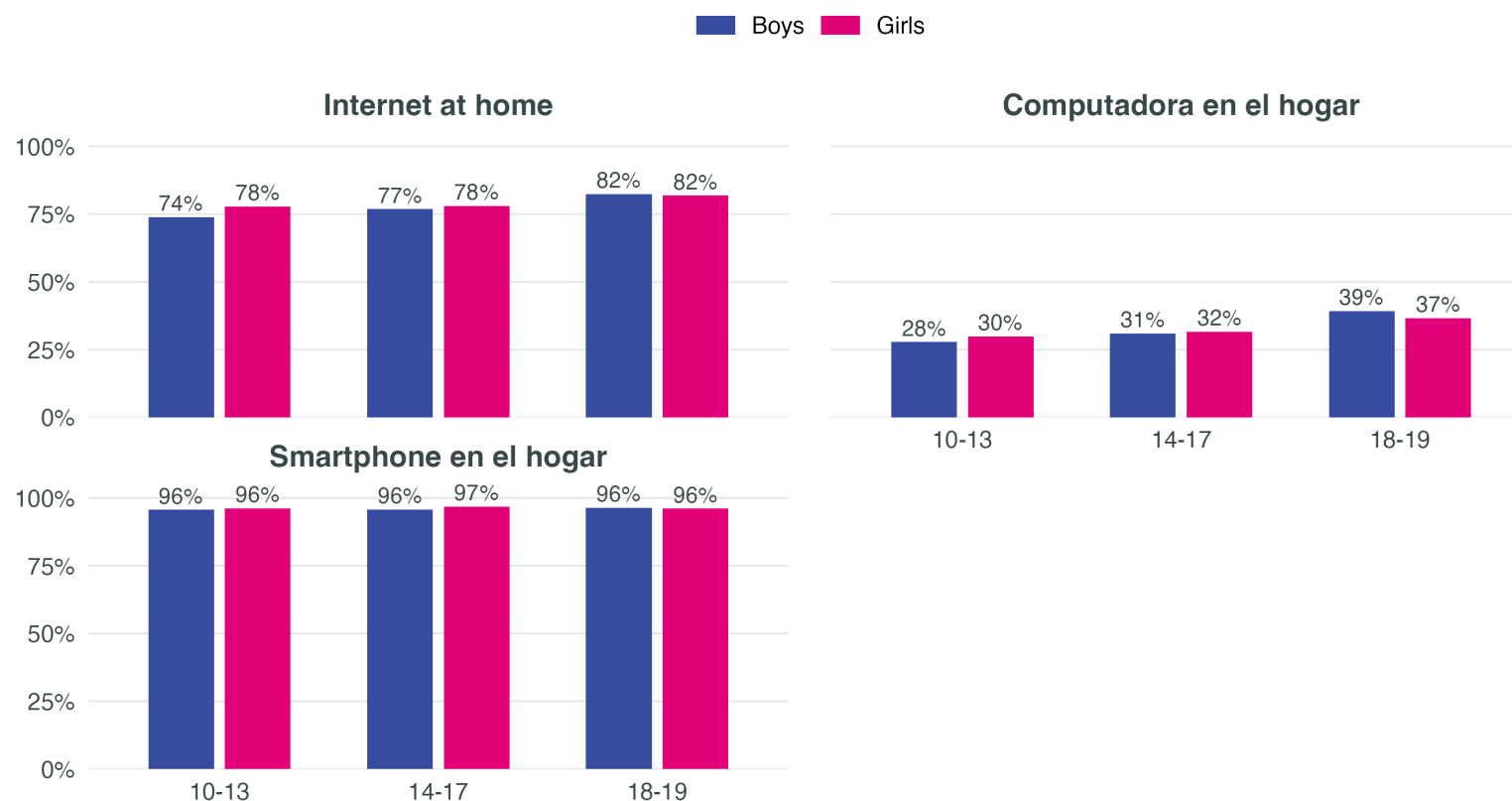


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.



# A22 · Household digital access — no differences by sex

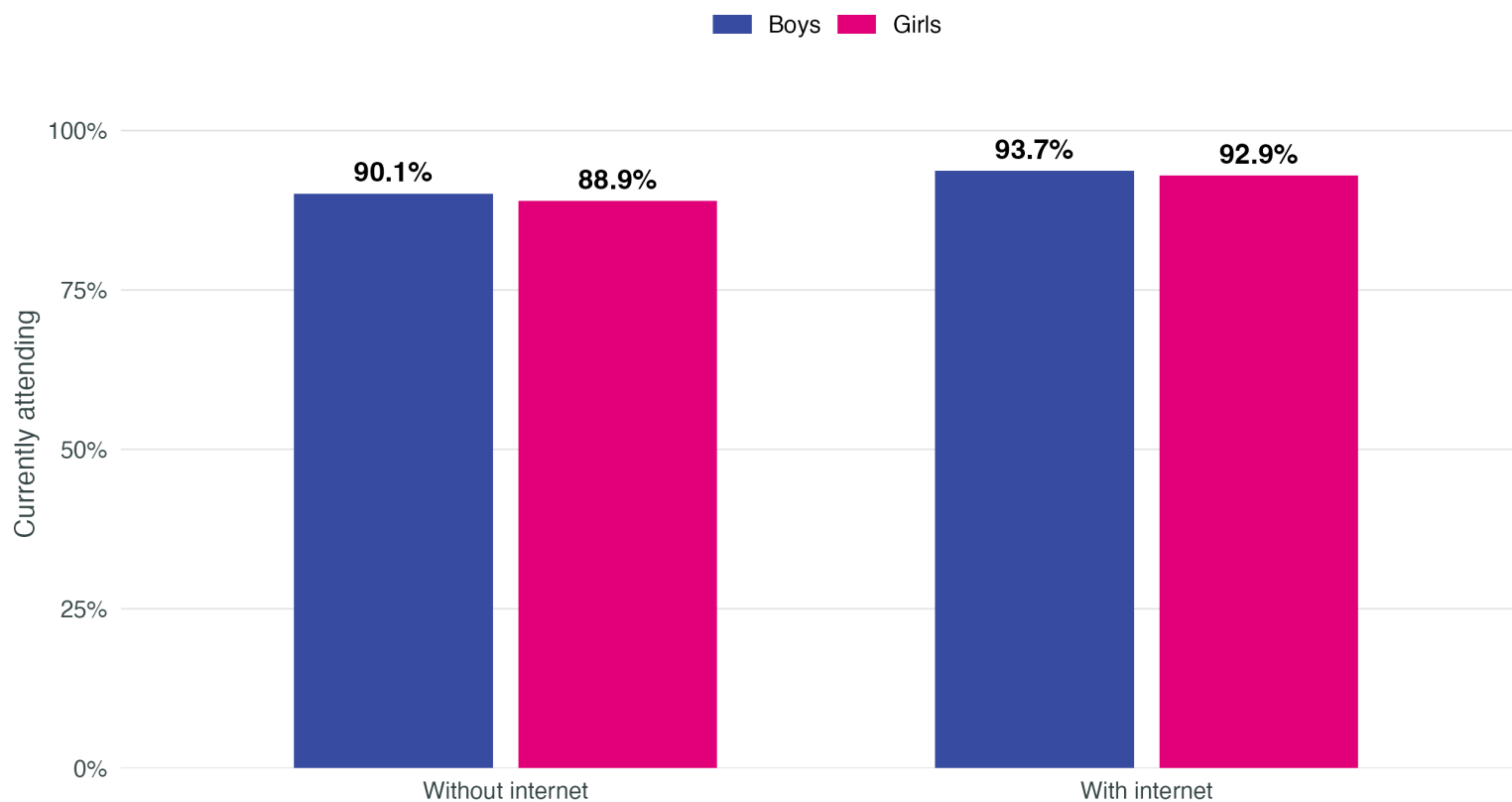
Share of girls and boys in households with each digital asset, by age group



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

# A23 · Connected households and girls' attendance

School attendance rate of adolescents 10–19, by household connectivity



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

**Note:** correlation, not causation. Households with internet also tend to have higher income, urban location, and parental education.